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Temperature-dependent current conduction properties and barrier inhomogeneity of Au/methylene blue (MB)/n-Ge heterostructure

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ABSTRACT

The electrical properties of Au/methylene blue (MB)/n-Ge heterostructure were investigated in a wide temperature range from 125 to 400 K. The device parameters such as barrier height, ideality factor and series resistance were determined using the thermionic emission (TE) model and Cheung's method. The barrier height (Φ_b) and ideality factor (n) values of the Schottky contact were determined from the current–voltage (I–V) measurements and found to be 0.29 eV and 2.71 at 125 K and 0.93 eV and 1.04 at 400 K, respectively. In the presence of inhomogeneity at the metal–semiconductor contact, the barrier height was found to be decreased and the ideality factor increased with the decrease of temperature. From Cheung's plot, the series resistance (R_s) was found to be reduced with the increase in temperature. Barrier inhomogeneity has been elucidated using the thermionic emission theory based on the assumption of Gaussian distribution of barrier heights. However, the divergence in Schottky barrier heights of Au/MB/n-Ge heterostructure evaluated from I–V measurements indicates deviation from the TE theory. The conventional Richardson plot between $\ln(I_o/T^2)$ vs. $1000/T$ gives an activation energy of 0.31 eV and Richardson constant (A^*) of 1.14×10^{-9} Acm^{-2} K^{-2}. The modified Richardson plot evaluated by assuming the Gaussian distribution of Φ_b shows an enhanced activation energy of 1.15 eV and A^* of 209.28 Acm^{-2} K^{-2} which is close to the theoretical value of n-Ge. Current conduction mechanisms of the Au/MB/n-Ge contact in a wide temperature range are resolved into four linear regions (Region-I to Region-IV) with different slope factors. This shows that the interfacial layer (MB) significantly influences the electrical properties of the Au/n-Ge contacts measured in a wide temperature range. The interface state density distribution over the energy below the conduction band of the n-Ge is also studied in the temperature range from 125 to 400 K.

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1 Introduction

Over the last few years, understanding the properties of organic materials such as electrical conductivity, sensing and high efficient luminescence properties has made them used in the fabrication of various hybrid optoelectronic devices such as solar cells, LEDs and thin film transistors. A very wide range of organic materials are available with π-conjugated arrangements of small molecules and polymers. The organic materials are lagging behind the inorganic materials given their inferiority in the mobility of carriers, charge carrier injection and environmental stability [1]. But, the advantage of high absorption coefficients, strong wave function overlap between HOMO and LUMO energy levels, simple solution processing techniques and customization of molecules in the organic materials makes these materials used in a wide range of advanced optoelectronic and photovoltaic applications [2–4]. The idea of making use of intrinsic properties of both organic and inorganic materials, the organic–inorganic heterojunctions will provide a brand-new capability of the hybrid system and expand the possibilities of isolated materials toward innovative and multifunctional applications [5–8].

A lot of efforts have been made to form thin films of organic materials using different fabrication procedures varying from low-cost spin coating to vacuum deposition. Notably, significant work was carried out on the fabrication of organic thin films on inorganic materials due to the advantage of the high mobility property of inorganic materials. Furthermore, the formation of organic thin films on inorganic materials significantly changes the conduction properties of the heterojunction devices. Though the organic thin film is unreactive, it creates a dipole layer on the near surface of the inorganic semiconductor which generally enhances the electrical properties of the hybrid devices. The Schottky contact formation to organic–inorganic hybrid devices shows its applicability in different areas ranging from photovoltaic devices to optoelectronic devices. The Schottky barrier parameters such as barrier height, ideality factors and series resistance are sensitive to various parameters such as interfacial layer, temperature, applied bias, distribution of localized interface states and barrier homogeneity. Further, the electrical measurements of organic–inorganic hybrid devices at room temperature don't reveal significantly the conduction mechanisms across the interfaces. Further, over the last two decades a lot of reports have evidenced the variation of Schottky barrier parameters of organic–inorganic hybrid devices with temperature. Hence, it is very important to investigate the current conduction mechanisms of hybrid devices over a wide temperature range.

The electrical properties of the Schottky contacts are generally explained based on thermionic emission current transport mechanism which can also be utilized to evaluate different barrier parameters such as barrier height, ideality factor and series resistance. But often these barrier parameters are dependent on temperature and defects caused at the interface due to inhomogeneity. In the case of temperature-dependent electrical properties of the contacts, the barrier parameters specifically barrier height and ideality factor deviate from ideality and are not well fitted with the thermionic emission model. The deviation in barrier parameters from the thermionic emission model might be explained based on barrier inhomogeneity caused by potential fluctuations at the interface. These potential fluctuations may transport the thermally activated carriers from low barrier height patches at low temperatures and high barrier height patches at high temperatures. Thereby the distribution of potential fluctuations across the interface might be varied based on the surface preparations, annealing temperatures, annealing ambience and also organic or inorganic interfacial layers used. Over the last two decades, investigators have explored different analytical models to explain the temperature-dependent electrical properties of the contacts including T_o effect proposed by Padovani and Sumner [9], the Gaussian distribution of barrier heights [10], potential fluctuation model proposed by Werner and Guttler [11], flat band barrier height (zero field) [12] and more widely recognized double Gaussian distribution barrier inhomogeneity model proposed by Tung [13].

Organic dyes are environmental contaminants which generally be produced from sources such as textile, paper and plastic industries. These compounds prevent sunlight permeation into the water, decrease the photosynthetic action and also cause the death of aquatics. In general, some of the dyes contain known carcinogenic aromatic groups. Methylene blue (MB) is a heterocyclic dye with harmful and toxic effects on water. Photocatalytic methods are introduced as green and low-cost approaches for the effective elimination of dyes from wastewater under mild conditions. Recently these drawbacks have been overcome to use these MB organic dyes in an energy conversion system as photosensitizers which shows absorption of light at 660 nm and also had lot of applications in medicine [14, 15]. Few reports are available taking advantage of its organic dyes and fabricated organic–inorganic hybrid devices using these dyes and studying its electrical and photovoltaic properties [16–20].

Temperature-dependent electrical measurements of metal-Ge Schottky contacts were reviewed over the past decade by many researchers [21–27]. The electrical properties of metal (Sn, Al, Au)/Ge Schottky contacts were analyzed by varying the temperature from 111 to 300 K [21]. The electrical properties reflect pure thermionic emission which is the dominant current transport phenomena observed in these contacts. Pitale et al. [22] investigated the temperature and bias-dependent electrical properties of Al/Ge (both p-type and n-type) contacts over a wide temperature range from 70 to 300 K. They analyzed the influence of charge neutrality level (CNL) on the net carrier concentration of the Al/Ge contacts with temperature. Low-temperature electrical properties were analyzed by presuming the inhomogeneity in barrier height distribution at NiGe/n-Ge contacts [23]. Ulusan and Tataroglu [24] fabricated AuGe/n-Ge devices and analyzed current–voltage properties in a broad temperature range from 200 to 400 K. They found a strong dependence of barrier parameters with temperature and also suggest the presence of barrier inhomogeneity at the metal-Ge interface. The influence of annealing on the electronic properties and low-temperature charge transport processes of Pt/n-Ge contacts was analyzed by Guo et al. [25]. They reported that the statistical barrier parameters and charge transport processes with device measuring temperature (183 K to 303 K) were significantly changed concerning the annealing temperature of the contacts. The annealing temperature creates a thin interlayer and the formation of metal-Germanides which changes the density of surface states, traps and reduces non-stoichiometric defects at the Pt/Ge interface. Current conduction processes and Schottky barrier parameters with temperature were explored in Se-modified n-Ge contacts [26]. The barrier parameter shows usual Gaussian barrier height distribution across the interface. Further, they also observed the Schottky emission is the foremost mechanism in the charge transport process at low temperatures. A similar type of barrier inhomogeneous characteristics was observed in resistive evaporated Au/Ge contacts [27].

Few reports are available to understand the current transport processes evolved in n-Ge contacts modified by various interlayers. Petit et al. [28] studied the very low temperature-based electronic measurements from 30 K to 300 K of $Mn_5Ge_3C_{0.6}$ layer modified doped and undoped n-Ge structures and analyzed the barrier properties and charge transport processes with temperature by assuming the standard thermionic model with the distribution of barrier height patches at the interface. Current–Voltage–Temperature (*I*–*V*–*T*) properties of permalloy-modified Ge contacts were analyzed and observed the variation of barrier parameters which proved the thermionic emission behavior modified with inhomogeneous barrier height [29]. Janardhanam et al. [30] also reported the conduction processes of WSe_2/p-Ge heterostructures in a wide temperature range and applied potential. They observed that the conduction process above 250 K is truly due to thermionic emission and below 250 K tunneling is found to be the dominant transport mechanism.

Very few works were acknowledged on temperature-influenced organic layer-modified n-Ge contacts. It was well known that the electronic parameters strongly depend on the organic layer between the metal–semiconductor contacts. Yildirim et al. [31] investigated the rubrene interlayer effects on barrier parameters of Au/n-Ge Schottky barrier diode at low temperatures. The temperature-influenced barrier parameters exhibit an inhomogeneous barrier and deviate from the thermionic emission model. Graphene-modified Ge-Schottky contacts were realized and studied its *I*–*V*–*T* features with and without graphene interlayer from 180 to 340 K. Abnormality in barrier parameters with temperature is observed in both the contacts due to inhomogeneous barrier height [32, 33]. Further, the lower barrier height distribution in Au/Graphene/n-Ge contacts is evidenced than in contacts without a graphene layer. This may be attributed to the reduction of native defects and passivation of the Ge-surface by the graphene layer. Understanding the current transport mechanism is vital for predicting the performance of the metal–semiconductor or metal/polymer/semiconductor junctions in various conditions and environments at varying temperatures. Temperature-dependent measurements also help to assess the thermal stability of the junctions, ensuring they can handle high-power applications without failure due to overheating. Heterojunctions are crucial in sensors and detectors applications which include radiation detection, light detection and gas sensing. *I*–*V*–*T* characteristics allow researchers to optimize the device sensitivity and responsivity under different operating conditions (temperature, voltage). These measurements enable the extraction of the thermal activation energy of the barrier, providing insights into the quality of the junction and how it evolves with changes in environmental conditions like temperature. These measurements provide a path for deep understanding of charge carrier dynamics at the interface and guide the design of devices that are optimized for a wide range of operating temperatures.

Despite the importance of Ge-based devices in optoelectronic or photovoltaic applications, it is very much essential to investigate the current transport and barrier properties of the devices at low and high temperatures. Further, the organic thin layer on inorganic semiconductors invariably contributed to the change in current conduction properties of the contacts. In the present study, Methylene blue (MB) dye was spin-coated on the cleaned Ge wafer and its electronic properties over a wide range of temperatures from 125 to 425 K in steps of 25 K. Classical thermionic emission (TE) model is applied to analyze the barrier properties of Au/MB/n-Ge contacts for the temperatures from 125 to 425 K. The deviation of barrier properties such as barrier height and ideality factor from the TE model was explained with well-accepted Tung's model of Gaussian distribution of barrier height. A detailed understanding of the conduction mechanisms of contacts was explored with temperature and bias potential. Dispersion of capacitance observed at elevated temperature indicates the temperature-dependent current transport processes influenced by series resistance and interface states.

2 Experimental

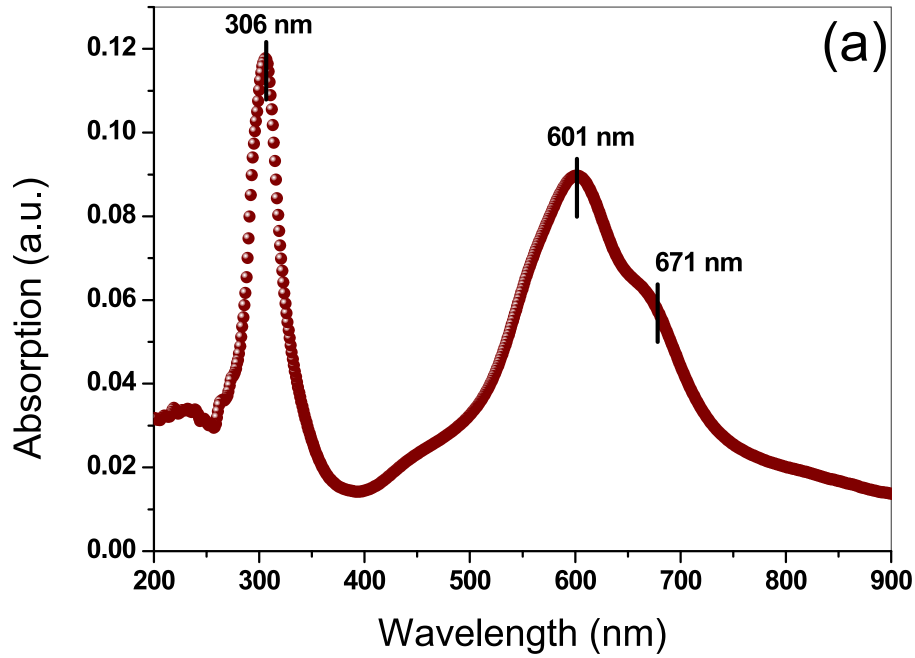
In the current study, an n-type Ge wafer (100) (Sb-doped) wafer with (100) orientation was employed. Firstly, the Ge-wafer was cleaned in trichloroethylene, acetone and methanol (in pure form as supplied by the manufacturer) each with 5 min. using ultrasonic agitation. Subsequently, it was etched by H_2O: HF (100:1) to remove the native oxide layer. After each stage of cleaning, the wafer was rinsed in deionized water and dried in N_2 atmosphere. Then the samples were immediately transformed into a vacuum coating unit to deposit ohmic contact (Al) of 50 nm thickness on the rough side of the sample. Methylene blue (MB) dye (in a powder form) purchased from Sigma-Aldrich with a dye content ≥ 82% and dissolved in water at 1 mg/ml concentration. The prepared solution is spin-coated on the cleaned n-Ge wafer at 4000 rpm for 1 min. with subsequent heating at 150 °C for 5 min. and the thickness of the MB dye was identified as ~ 50 nm using a profilometer. Using profilometer, a step height is measured from uncoated surface to MB coated surface which determines the thickness of the MB thin film. An average of several measurements (nearly 10 measurements) was utilized to determine the thickness of MB thin film. Then Schottky contact (Au) was e-beam evaporated on the top of the MB layer using a shadow mask of 0.5 mm diameter. The thickness of the Au contact was found to be 40 nm measured using a digital thickness meter. All vacuum evaporations were performed at a base pressure of 5×10^{-6} mbar. Surface morphology of MB thin films coated on n-Ge substrate was evaluated using AFM measurements in tapping mode (Nanosurf AFM). Optical studies were also carried out for MB films on glass plates using UV–Vis measurements (JASCO, V-670 model).The electrical properties of the Au/MB/n-Ge contacts were performed at temperatures varied from 125 to 400 K with a step of 25 K (using a precision semiconductor parameter analyzer (Agilent 4156C).

3 Results and discussion

3.1 Optical and temperature-dependent electrical parameters

Figure 1 shows the morphological and optical spectra of MB thin films. The morphology of MB thin films from AFM measurements in the surface area of 3 μm × 3 μm reveals a very smooth surface with RMS roughness of 6.17 nm. The optical absorption measurements of MB films on glass plates show three absorption peaks at 306 nm, 601 nm and 671 nm. These absorption peaks are well matched with the literature and absorption peaks observed typically at 306 nm correspond to $\pi \rightarrow \pi^*$ transitions [34, 35].The absorption peaks of MB thin films are well correlated with the absorption studies carried out for MB solution (the graph is not shown here).

Figure 2 reveals the reverse and forward current–voltage (*I*–*V*) properties of Au/methylene blue (MB)/n-Ge Schottky contact measured at



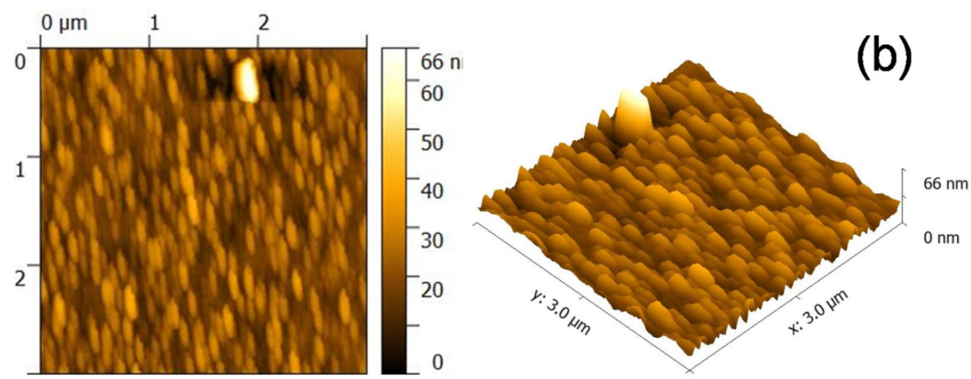


Fig. 1 **a** Absorption spectra of methylene blue thin films on glass plate **b** AFM microstructure of methylene blue thin films on Ge substrate

temperatures from 125 to 400 K in steps of 25 K. The observed *I*–*V* properties of the contact and the evaluated Schottky barrier parameters based on standard thermionic emission (TE) theory [36] shows a strong dependence on measurement temperature. The Schottky barrier parameters such as barrier height (Φ_b) and ideality factor (n) were evaluated from the linear characteristics of the forward bias *I*–*V* graph. The variation of extracted magnitudes of n and Φ_b of the MB/Ge contact against the measurement temperature is plotted in Fig. 3a and is mentioned in Table 1. The Φ_b of MB/Ge contact varied from 0.29 eV at 125 K to 0.93 eV at 400 K and the magnitude of n varied from 2.71 at 125 K to 1.04 at 400 K. This variation specifies that the deviation of the TE model at low temperatures and also the decrease in the magnitude of n and increase in Φ_b with increasing in temperature, suggests the presence of inhomogeneities at the interface. Moreover, the presence of MB thin film between the metal and semiconductor can affect the inhomogeneity of the Schottky barrier height and also affect the current conduction through the metal/semiconductor (MS) junction. The current transport mechanism across the MS junction is a temperature-dependent process where the electrons are transported through a low barrier height at the

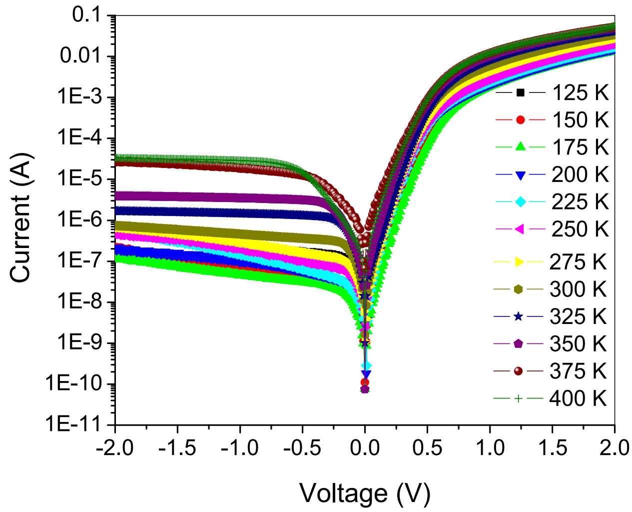


Fig. 2 Temperature-dependent *I–V* measurements of Au/Methylene Blue (MB)/n-Ge SBDs in the temperature range of 125–400 K

lower temperature and a high barrier height at the higher temperature. The larger magnitudes of n at low temperatures suggest the wide spread of low barrier height regions, prevails tunneling process via the interfacial layer and an inhomogeneous barrier [37]. The obtained range of Φ_b at different measurement temperatures signifies potential fluctuations at the interface which are generally caused due to non-uniform interfacial charges, defects and interface layer composition [32]. These observed variations of Φ_b and n with measurement temperature are consistent with the earlier reports in the case of interlayer/n-Ge contacts [29, 31, 32].

The series resistance (R_S) of the MB/n-Ge contact found its importance at the non-linear or curved region of the *I–V* graph which influences the Φ_b and n magnitudes.Fig. 3b depicts the variation of series resistance with temperature determined from d*V*/dlnI vs. *I* and *H*(*I*) vs. *I* graph. To determine the Schottky parameters such as ideality factor (*n*), barrier height and series resistance (R_S) by using the temperature-dependent *I–V* measurements, Cheung's method was adopted [15]. The extracted R_S and n magnitudes are found to be 80 Ω and 6.71 at 125 K and 29 Ω and 2.14 at 400 K, respectively. From the estimated ideality factor using d*V*/dlnI vs. *I* graph, another Cheung's function *H*(*I*) was utilized to evaluate the series resistance and barrier height [31]. The extracted R_S and barrier height

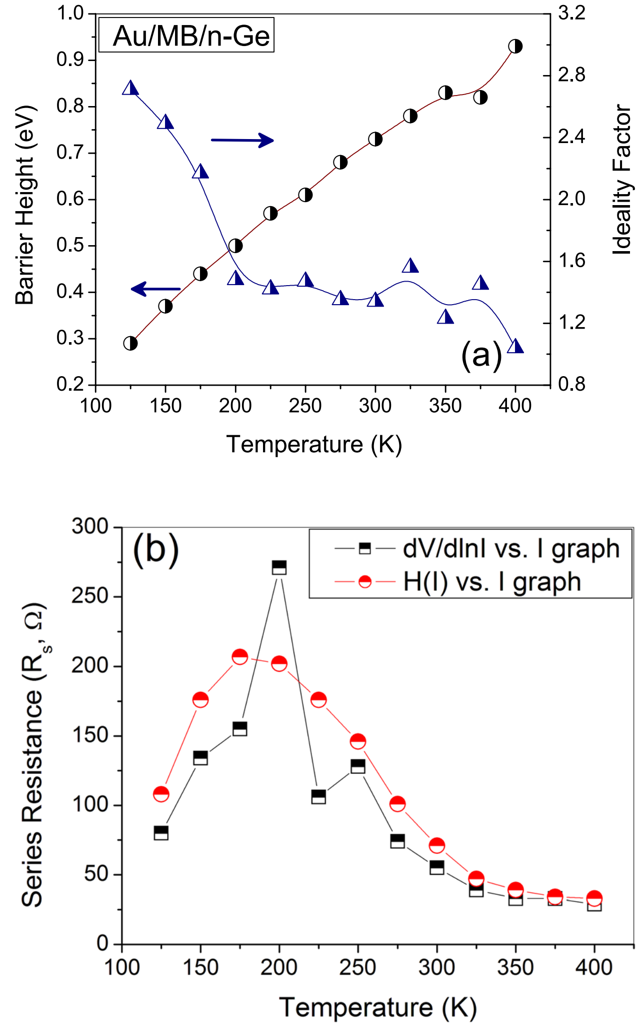


Fig. 3 **a** Temperature dependence of ideality factor (*n*) and barrier height (Φ_b) of Au/MB/*n*-Ge SBDs. **b** Variation of series resistance with temperature determined from d*V*/dlnI vs *I* and *H*(*I*) vs I

(Φ_b) magnitudes obtained from the *H*(*I*) vs. *I* plot are found to be 108 Ω and 0.11 eV at 125 K and 33 Ω and 0.53 eV at 400 K, respectively. The evaluated R_S value from the d*V*/dlnI vs. *I* is almost identical to the plot of *H*(*I*) vs. *I*, this implies the consistency and validity of the TE model. Increased magnitude of R_S in both methods at 125 K can be due to a reduction in the carrier density and its mobility, which further increases the density of trapping centers filled with the carriers at the interface [38]. On the other hand, the decreased trend of R_S at high temperatures might be ascribed to increased density of charge carriers and also the carrier acquired enough thermal velocity which may escape from the traps.

Table 1 Temperature dependence of electrical parameters evaluated using *I–V* characteristics of Au/MB/n-Ge heterojunctions

Temperature (K)	Φ_b (eV)			n		R_S (Ω)		
	ln(*I*)-*V*	*H*(*I*)	Norde	ln(*I*)-*V*	d*V*/dlnI	d*V*/dlnI	*H*(*I*)	Norde
125	0.29	0.11	0.26	2.71	6.71	80	108	30,917
150	0.37	0.13	0.34	2.49	6.28	134	176	7061
175	0.44	0.18	0.36	2.17	4.87	155	207	37,687
200	0.5	0.16	0.42	1.48	5.44	271	202	58,174
225	0.57	0.06	0.46	1.42	13.74	106	176	38,898
250	0.61	0.21	0.5	1.47	4.44	128	146	14,596
275	0.68	0.25	0.51	1.35	3.87	74	101	3460
300	0.73	0.31	0.51	1.34	3.19	55	71	469
325	0.78	0.41	0.53	1.56	2.5	39	47	173
350	0.83	0.43	0.57	1.23	2.44	33	39	294
375	0.82	0.51	0.67	1.45	2.13	33	34	1225
400	0.93	0.53	0.62	1.04	2.14	29	33	115

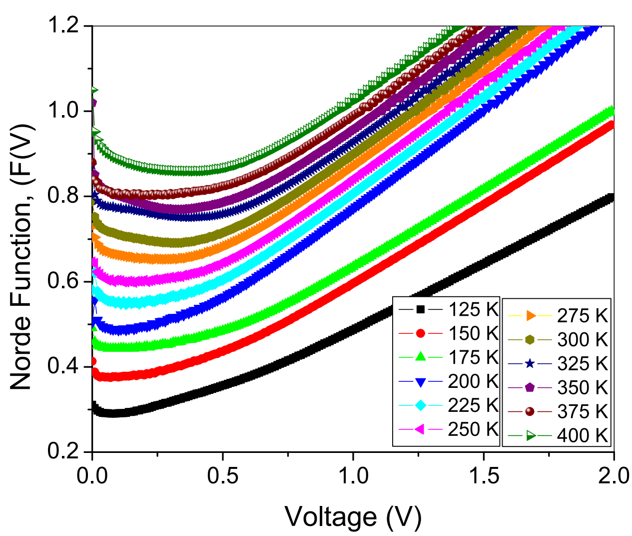


Fig. 4 Norde plot of Au/MB/n-Ge SBDs at elevated temperature

Norde's method can be employed for the evaluation of the series resistance and Φ_b from the forward bias *I–V* for ideal diodes devoid of interfacial disorder [39]. Norde function as defined in [39, 40] is utilized to determine the barrier parameters. A plot of Norde function concerning at forward bias voltage of MB/n-Ge junction in the temperature range of 125 to 400 K is presented in Fig. 4. The magnitude of Φ_b is found to be varied from 0.26 eV to 0.62 eV whereas the series resistance varied from 30.97-- kΩ to 115 Ω for the increase of temperature range from 125 to 400 K. The magnitudes of barrier height obtained from TE model, Cheung's function and Norde method are highly comparable. Moreover, the values of R_S obtained from the Norde function show its considerable deviation from the values of the same obtained from Cheung's method. This might be due to the regional dependence of the approaches. The entire region of *I–V* data is considered for the evaluation of R_S in the case of the Norde approach. On the other hand, a non-linear region is considered in Cheung's method for the evaluation of R_S. Further, the Norde method is not suitable for the contacts having a larger ideality factor which shows its inconsistency with pure TE theory [41].

Figure 5a shows the forward bias log (I) versus log (V) plot of Au/MB/n-Ge SBDs measured in the temperature range from 125 to 400 K. The forward bias log (I) versus log (V) curve is characterized by distinct linear regions that can indicate different conduction mechanisms based on the slopes of the linear regions. The forward bias measurements of MB/n-Ge heterostructure at varied temperatures from 125 to 400 K show four linear regions as evidenced in Fig. 5. The slope of the different linear segments (named as Region I, Region II, Region III and Region IV) is plotted against temperature for the increased bias range (shown in Fig. 5b). As shown from Fig. 5b the Region I (0.11 > V > 0.02) shows a linear segment with a slope varied from 1.94 to 1.57 for the increased temperature range from 125 to 400 K, respectively. The slopes evidenced in Region I define ohmic conduction as the dominant current transport process. Further, Region II is observed in the bias range from 0.11 to 0.34 V reveals the slope varied from 3.39 to 3.64 for the temperature range from 125 to 400 K, respectively. As the bias voltage is increased, the

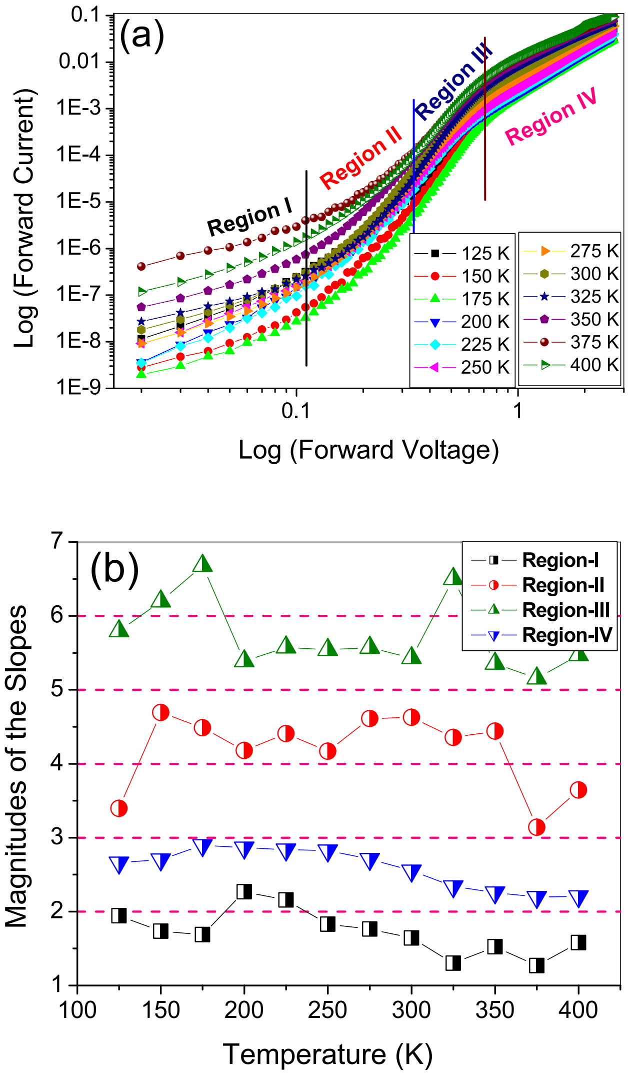


Fig. 5 **a** Forward bias log (I) vs. log (V) plot of Au/MB/n-Ge SBDs in the temperature range from 125 K–400 K **b** Magnitude of slopes obtained from linear segments of log (I) vs. log (V) plot with respect to temperature range from 125 K–400 K

slope of the linear segments seems to be greater than two suggests the space charge limited current (SCLC) conduction is dominated than the thermally induced carrier conduction. Similarly, Region III, which is seen in the bias range of 0.35 V to 0.7 V, was found to have linear segment slopes that varied from 5.79 to 5.47 for the temperature variations from 125 to 400 K. The larger magnitude of slopes much greater than the two observed in Region III might suggest the trap-assisted SCLC conduction process is the dominant transport process. Finally, in Region IV (0.7 V to 2.7 V), the magnitude of slopes varied from 2.66 to 2.2 for the temperature variation from 125 to 400 K. The magnitude of slopes observed at much larger bias voltages indicate the conduction process reaches to trap-filled limit and trap-free SCLC is the dominant conduction process [42, 43].

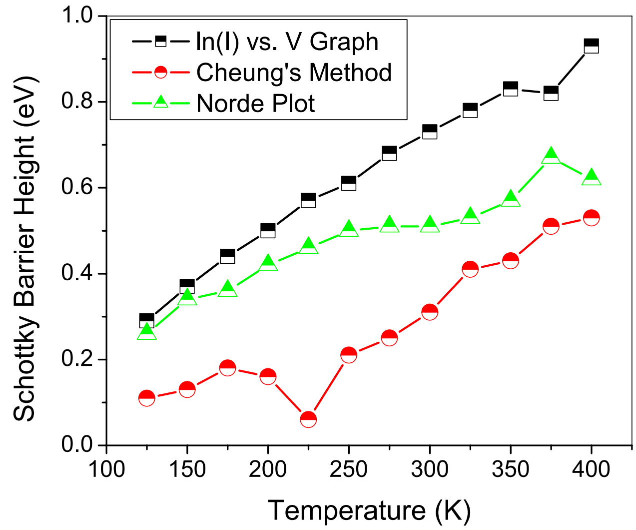


Fig. 6 Plots of barrier height vs. temperature obtained from the ln(I)–V, Cheung's and Norde methods

The temperature dependency of the barrier height determined using different approaches is shown in Fig. 6. The barrier height determined by the generalized Norde approach and those derived by the traditional *I*–*V* method correspond well, as illustrated in Fig. 6. The Norde technique necessitates that the contact between the metal–semiconductor is perfect (i.e., $n = 1$), which is why there is a discrepancy between it and the standard *I*–*V* approach. The temperature dependence of Schottky barrier height values obtained using the traditional *I*–*V* approach is identical to that of the same computed using Cheung functions. An increase in the barrier height and a decrease in the ideality factor of the MB/n-Ge heterojunction with temperature indicate the deviation of the thermionic emission mechanism. This behavior can be explained based on the potential fluctuation model for spatial inhomogeneous contacts by assuming the Gaussian distribution of barrier heights [44]. Since current transport across the MS interface is a temperature-activated process, electrons at low temperatures are able to surmount the lower barriers and therefore current transport will be dominated by current flowing through patches of the lower barrier height and a larger ideality factor. As the temperature increases, more and more electrons have sufficient energy to surmount the higher barrier. As a result, the dominant barrier height will increase with the temperature and bias voltage [11–13].

3.2 Properties of inhomogeneous barrier of MB/n-Ge heterostructure

The variation of barrier parameters such as barrier height and ideality factor of the MB/n-Ge heterostructure with temperature could be associated with the spatial distribution of barrier heights following Gaussian distribution [44]. Richardson plot is generally utilized to understand the influence of forward bias potential on the barrier height and also helps to evaluate the experimental determination of Richardson's constant [40]. Richardson plot can be analyzed by taking a variation of $\ln(I_o/T^2)$ vs. 1000/T as evidenced in Fig. 7a. A non-linearity was observed in the Richardson plot, the activation energy (E_a) and Richardson constant (A^*) were evaluated as 0.31 eV and 1.14×10^{-9} Acm^{-2} K^{-2} from slope and intercept of the plot respectively. The obtained values of the Richardson constant seem to be much less than the theoretical value of A^* of n-Ge [31]. These abnormalities could be addressed with barrier inhomogeneity and potential fluctuation models. In Schottky junctions, the Richardson constant plays a critical role in analyzing the current–voltage characteristics, where deviations from the ideal theoretical value can indicate issues like barrier height inhomogeneity. Richardson's constant significantly impacts the current conduction mechanism, particularly in thermionic emission. Different factors influence the low magnitude of Richardson constant obtained from temperature-dependent ideality factor. These factors include interface roughness, distribution of density of states, impurity levels and interface properties of MB-Ge interface. In metal–semiconductor junctions, it is truly the interface quality which can significantly influence the Richardson constant. Further, the modified Richardson plot by assuming Gaussian distribution of barrier heights at the MB-Ge interface gives a good straight line and also provides a better estimation of Richardson's constant, effective barrier height.

Understanding the potential fluctuation model and Gaussian distribution of Φ_b across the MB-Ge interface needs to evaluate zero-bias barrier height (Φ_{bo}) and standard deviation σ_o. These parameters are evaluated using a plot drawn between Φ_b versus

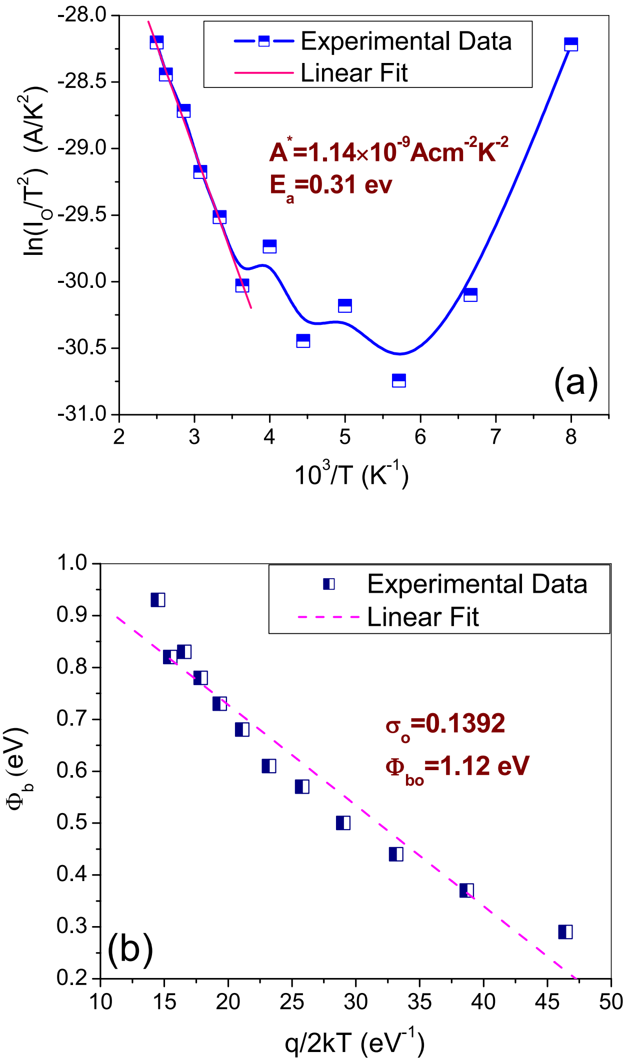


Fig. 7 Analysis of barrier inhomogeneity at MB-Ge interface in the temperature range of 125 K–400 K **a** Richardson plot drawn between $\ln(I_0/T^2)$ vs. $10^3/T$ **b** Variation of Barrier height vs. q/2kT

q/2kT as shown in Fig. 7b. The plot reveals a linearity between Φ_b and q/2kT with slope yielding σ_o and the intercept at $T = 0$ reveals Φ_{bo} based on the following expression [45],

$$\Phi_b = \Phi_{bo}(T = 0) - \frac{q\sigma_o^2}{2kT} \tag{1}$$

The plot between Φ_b and q/2kT reveals the σ_o and Φ_{bo} which were found to be 0.1392 and 1.12 eV, respectively. These parameters are utilized further to modify the conventional Richardson plot by combining the Eq. (1) with the following equations obtained from basic TE theory [46],

$$I_0 = AA^*T^2 \exp\left(-\frac{q\Phi_b}{kT}\right) \tag{2}$$

$$\ln\left(\frac{I_0}{T^2}\right) = \ln\left(AA^*\right) - \frac{q\Phi_b}{kT} \tag{3}$$

$$\ln\left(\frac{I_0}{T^2}\right) - \frac{q^2\sigma_0^2}{2k^2T^2} = \ln\left(AA^*\right) - \frac{q\Phi_{bo}}{kT} \tag{4}$$

The plot of the modified Richardson plot is illustrated in Fig. 8 reveals a linear relation. The slope and intercept of this plot disclose the modified Richardson constant (A^*) and the zero bias barrier height (Φ_{bo}), which was found to be 209 Acm^{-2} K^{-2} and 1.17 eV, respectively. The magnitude of Φ_{bo} obtained from Eq. (4) seems to be very well correlated with Eq. (1) suggesting the temperature-dependent electrical properties of MB/n-Ge heterostructure can be successfully addressed with TE theory modified with Gaussian distribution of barrier heights.

Dispersion of capacitance at 1 MHz with bias voltages under the influence of temperature is depicted in Fig. 9a. It is evident from the figure that the capacitance of the MB/n-Ge junction is dependent both on bias and temperature particularly in depletion and inversion regions. Region of the forward bias, which signifies the dependence of surface state density and series resistance [47]. The peak shift towards negative bias voltage with increasing temperature could be the indication of the reordering of the interface states and series resistance [48, 49]. A peak also observed in the depletion region of the C–V properties of the junction could also be associated with the presence of inhomogeneity at the MB-Ge junction [50]. As the carrier migration across the barrier is a temperature-activated process, the dispersion of capacitance at low temperatures in the reverse bias seems to be negligible as carriers get insignificant thermal energy. Whereas at elevated temperatures the migration of the density of carriers increases due to enhanced thermal energy and both the density of states and series resistance enhance the dispersion of capacitance. Similar temperature-dependent dispersion of capacitance of

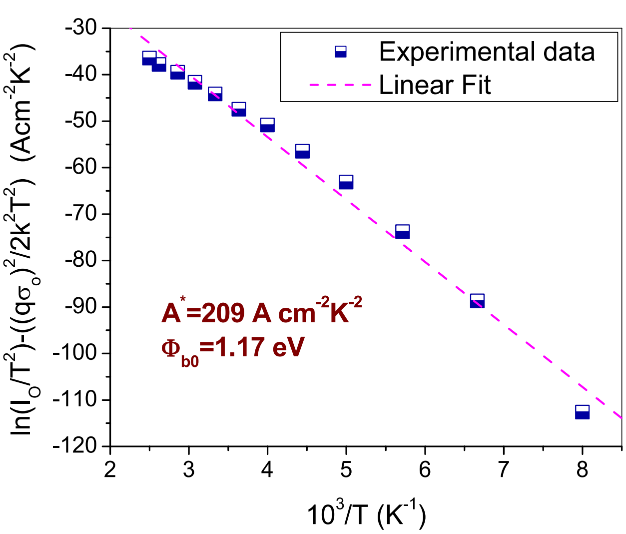


Fig. 8 Modified Richardson plot assuming TE modified by Gaussian distribution of barrier height in the temperature range of 125 K–400 K

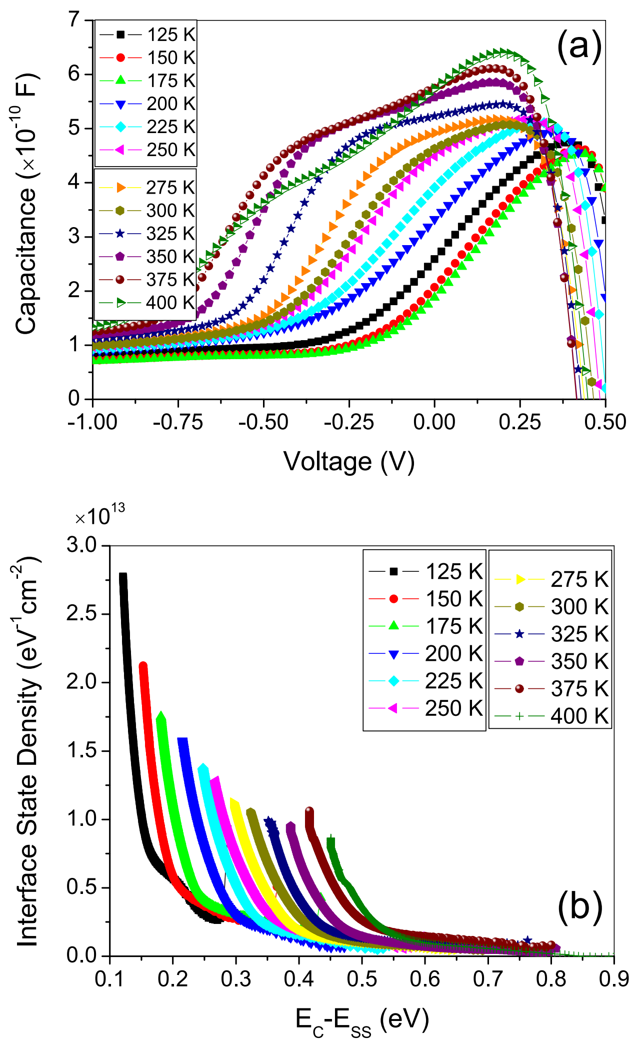


Fig. 9 **a** Junction capacitance vs. voltage of the heterostructure **b** Variation of density of states (N_{SS}) plotted against E_C–E_{SS} for the MB/n-Ge heterojunction in the temperature range from 125 to 400 K

heterojunctions/MOS devices is evidenced in earlier published works [47–50].

The distribution of the density of states present at the MB-Ge interface is also a temperature-activated process. Hence, understanding the temperature-dependent variation of the density of states at the MB-Ge interface is very important to support the current conduction mechanisms across the interface of the MB/n-Ge heterostructure. According to Card and Rhoderick [51], the interface state density (N_{SS}) is plotted against surface state energy (E_{SS}) below the conduction band as shown in Fig. 9b. The magnitude of N_{SS} expressed based on bias-dependent n ($n(V)$) as [51]

$$N_{ss}(V) = \frac{I}{q}\left[\frac{\varepsilon_i}{\delta}(n(V) - I) - \frac{\varepsilon_s}{W_D}\right] \tag{5}$$

On the other hand, in the case of an n-type semiconductor, the surface state energy (E_{SS}) below the conduction band is defined as,

$$E_c - E_{ss} = q\left(\Phi_b - V\right) \tag{6}$$

where δ is the thickness of the interlayer, ε_i and ε_s are the permittivity of the interlayer and n-Ge semiconductor and W_D is the width of the depletion layer. From Fig. 9b, it is evidenced that an increase in the magnitude of N_{SS} follows an exponential behavior and is found to be increased from the mid-gap of the semiconductor to the bottom of the conduction band. This possibly suggests the variation of barrier parameters with temperature associated with the particular distribution of density of states [52, 53].

Capacitance measurements are essential in understanding charge distribution and interface states in semiconductor devices. The capacitance dispersion observed in both the depletion and inversion regions can be attributed to the temperature-dependent variation of the interface state density (N_{SS}). At low temperature (125 K), the depletion region capacitance is significantly influenced by frozen-out carriers, leading to lower charge availability. The inversion region capacitance is suppressed because thermally excited carriers are not sufficient to form a strong inversion layer. The interface states closer to the conduction band contribute more significantly, leading to an increase in N_{SS} from 0.12×10^{13} eV^{-1} cm^{-2} to 3.41×10^{12} eV^{-1} cm^{-2} as E_C–E_{SS} decreases from 0.12 eV to 0.28 eV. But in the case of high temperatures (400 K), the depletion region capacitance increases due to enhanced carrier excitation, leading to more free carriers contributing to charge storage. The inversion capacitance also increases as more carriers participate in charge accumulation due to the thermal activation of deep-level states. The interface state density (N_{SS}) shifts, showing values from 8.66×10^{12} eV^{-1} cm^{-2} to 7.33×10^{11} eV^{-1} cm^{-2} for E_C–E_{SS} between 0.45 eV and 0.65 eV. These observations indicate that temperature directly modulates the depletion width and inversion charge dynamics, thereby impacting the capacitance behavior. The increase in N_{SS} near the conduction band edge suggests that interface states significantly impact carrier recombination and charge trapping, leading to an increased leakage current at higher temperatures, reduced device efficiency due to trap-assisted conduction and shifting of capacitance–voltage characteristics with temperature. Another reason of such behavior of capacitance with temperature may be the re-structuring and re-ordering of charges at surface and interface traps. In other words, it is believed that the trap charges have enough energy to evacuate the traps located between metal and semiconductor interface in the Au/MB/n-Ge bandgap at high temperatures [54].

4 Conclusions

The electrical properties of Au/methylene blue (MB)/n-Ge Schottky contacts were investigated in a wide temperature range from 125 to 400 K. The device parameters such as barrier height, ideality factor and series resistance were determined using the thermionic emission (TE) model and Cheung's method. The barrier height and ideality factor (n) values of the Schottky contact were determined from the current–voltage measurements and found to be 0.29 eV and 2.71 at 125 K and 0.93 eV and 1.04 at 400 K, respectively. In the presence of inhomogeneity at the metal–semiconductor contact, the barrier height was found to be decreased and the ideality factor was found to be increased with the decrease of temperature. From Cheung's plot, the series resistance (R_S) was found to be decreased with the increase in temperature. The barrier inhomogeneity has been elucidated using the TE theory based on the assumption of Gaussian distribution of Schottky barrier heights. However, the divergence in Schottky barrier heights of the Au/MB/n-Ge Schottky contacts evaluated from current–voltage measurements indicates the deviation from the TE theory. The conventional Richardson plot between $\ln(I_o/T^2)$ vs. 1000/T gives an activation energy of 0.31 eV and Richardson constant (A^*) of 1.14×10^{-9} $Acm^{-2} K^{-2}$. The modified Richardson plot evaluated by assuming the Gaussian distribution of the barrier shows an enhanced activation energy of 1.15 eV and Richardson constant (A^*) of 209.28 $Acm^{-2} K^{-2}$, which is close to the theoretical value of n-Ge. Current conduction mechanisms of the Au/MB/n-Ge contact in a wide temperature range are resolved into four linear regions (Region-I to Region-IV) with different slope factors. This shows that the interfacial layer (MB) significantly influences the electrical properties of the Au/n-Ge contacts measured in a wide temperature range. The density of states of Au/MB/n-Ge contact shows an exponential increase from the mid-gap of the semiconductor to the bottom of the conduction band.

Author contributions

Conceptualization, Mallikarjuna D and Ashok Kumar A; Data curation, Ashok Kumar A and Rajagopal Reddy V; Formal analysis, Janardhanam V and Ashok Kumar A; Methodology, Mallikarjuna D, Ashok Kumar A and Rajagopal Reddy V; Resources, Janardhanam V and Rajagopal Reddy V; Supervision, Ashok Kumar A; Validation, Ashok Kumar A, Janardhanam V and Rajagopal Reddy V; Visualization, Mallikarjuna D and Ashok Kumar A; Writing-original draft, Mallikarjuna D and Ashok Kumar A; Writing-review & editing, Ashok Kumar A, Janardhanam V and Rajagopal Reddy V. All authors have read and agreed to the published version of the manuscript.

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Data availability

Where no new data were created, or where data are unavailable due to privacy or ethical restrictions. The authors are herewith giving the consent that the data be made available on request.

Declarations

Conflicts of interest The authors confirm that there is no conflict of interest or competing interest about this work with anybody.

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